

Quantifying variations of thigh muscle geometries through automated CT segmentation

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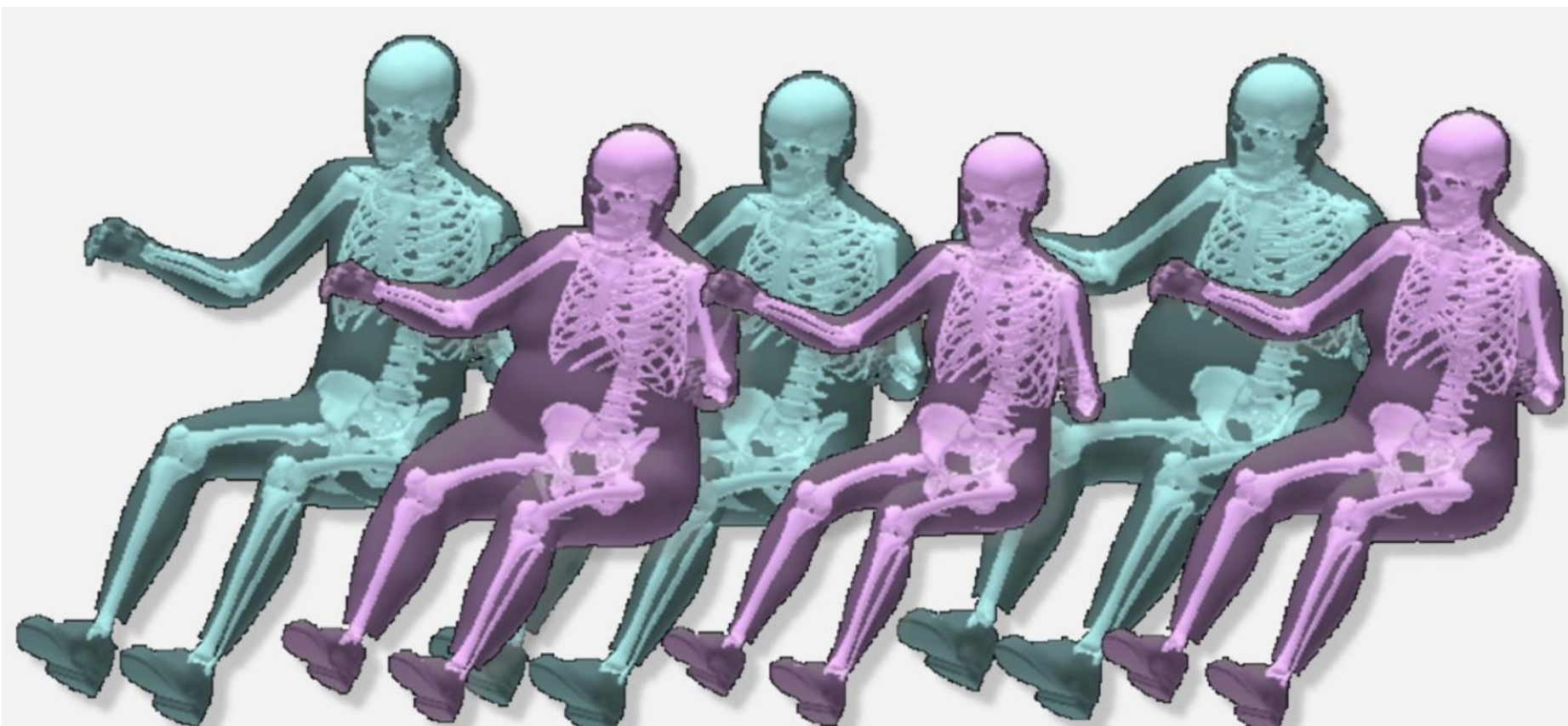
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Background

Human Anatomical Variability:
Demographic factors (age, sex, stature, BMI) alter occupant kinematics, seatbelt fit, and tissue injury tolerances during crashes.

Limitations of Traditional Models:
Existing HBMs and crash dummies rely on a few discrete sizes (e.g., 50th-percentile male), leaving continuous population variability underrepresented.

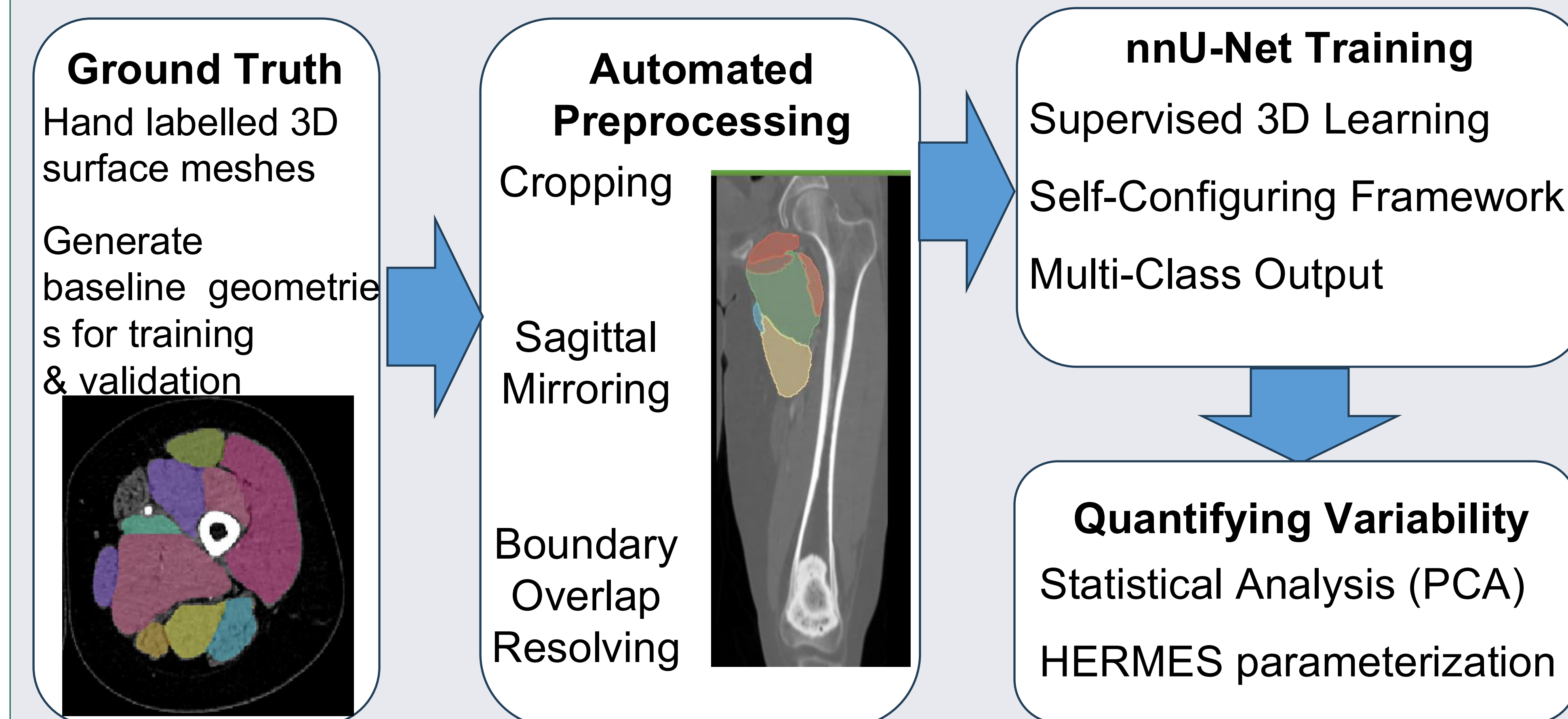
The HERMES Framework:
Developed at UMTRI, HERMES uses statistical shape models and mesh morphing to dynamically generate full-body digital twins across diverse demographics.



Objective

To establish a high-quality 3D ground-truth dataset and an automated preprocessing pipeline, enabling AI-driven auto-segmentation of individual lower-extremity muscles to **quantify human variability** for HERMES.

Methods

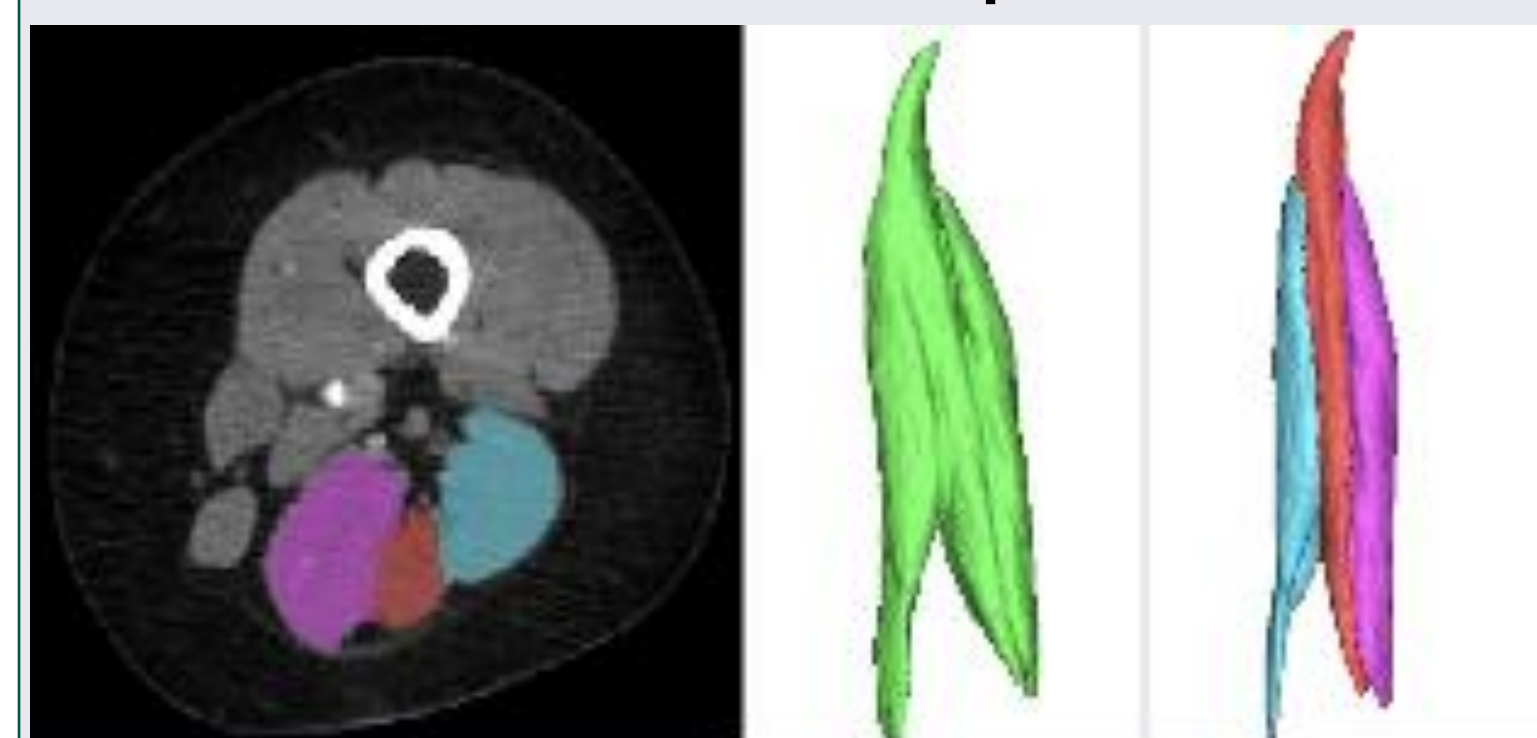


Results

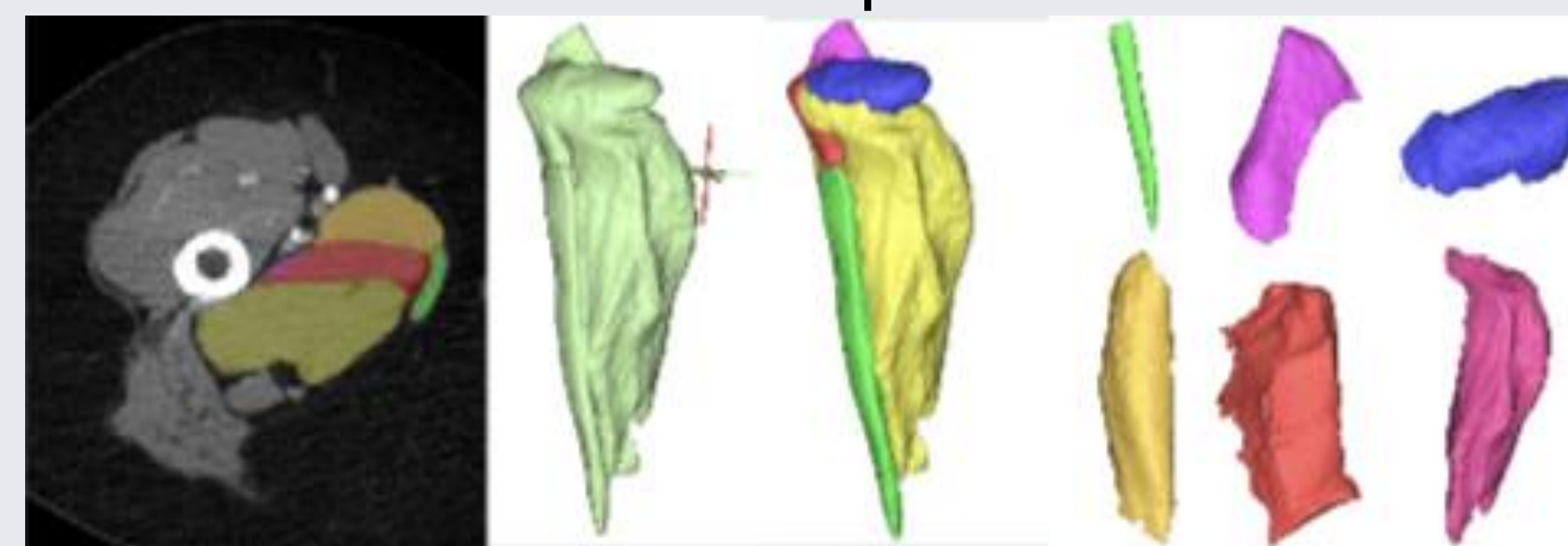
Dataset Yield: Successfully preprocessed and formatted **45 clinical CT scans** into standardized left/right leg volumes ready for nnU-Net ingestion.

Posterior and Medial Compartments

Posterior Compartment

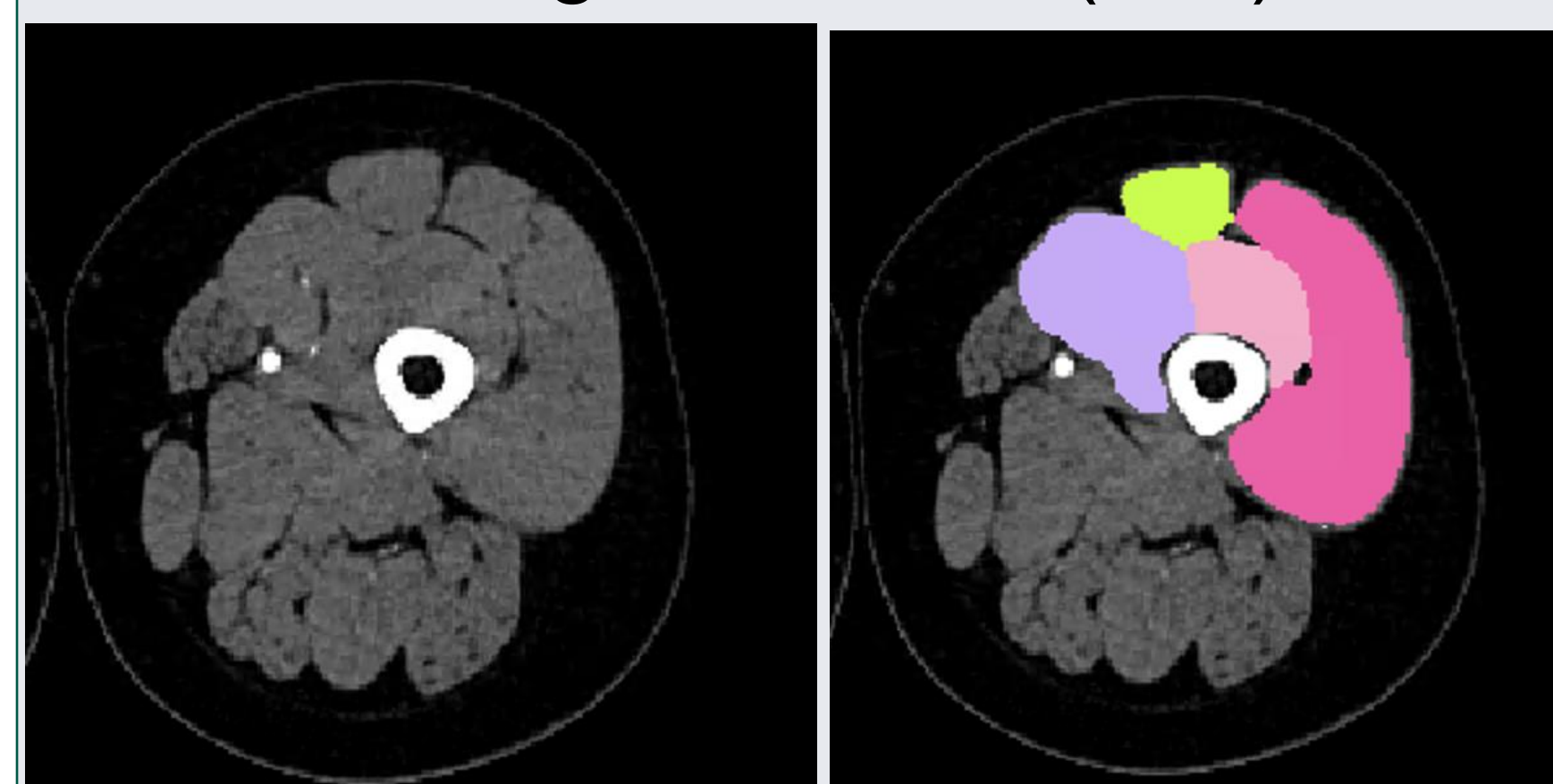


Medial Compartment

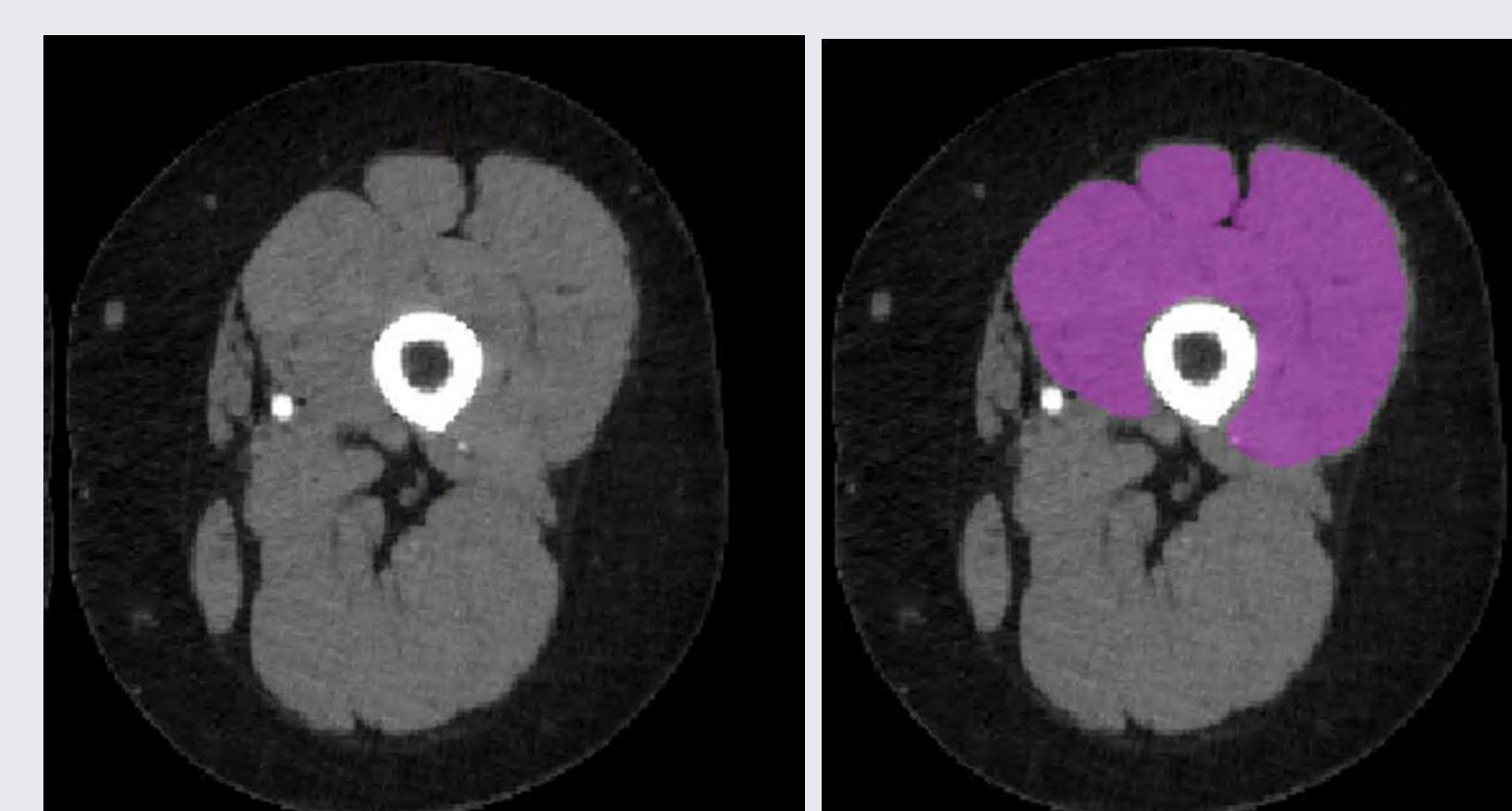


Quadriceps Femoris: Low intra-muscular contrast on standard clinical CTs, makes individual separation of the vastus group unreliable for most cases

High Contrast (few)



Low Contrast



Discussion and Impacts

Standardized Pipeline: Automated cropping, mirroring, and overlap resolution eliminate tissue clipping and ensure perfect spatial alignment for AI dataset staging.

Individual Muscle Isolation: Compartment lumping loses distinct pulling directions; isolating individual muscles preserves muscle orientation and cross-sectional areas critical for active HERMES crash modeling.

Contrast Divide & Bottleneck: Clinical CTs provide clear contrast for medial and posterior muscles, but low intra-quadriceps contrast obscures boundaries, requiring future statistical shape priors (PCA)

Next Steps

AI-Assisted Data Generation: Deploy initial baseline model to run semi-automated inference, scaling the dataset to 50 cases with manual fine-tuning.

Statistical Geometry Modeling: Perform statistical shape analysis (e.g., PCA/regression) on isolated muscle geometries to quantify anatomical variations for HERMES parameterization.

Conclusion

This automated pipeline standardizes raw CTs into ML-ready datasets while preserving muscle orientations.